# Answers and talk notes

Prime Factorization

### What your student is learning

Tonight we build a factor tree all the way down to primes — and learn to tell when the tree is actually finished.

## Answers

Ask your student to explain before you look.

1. **A.** 2, 2, 3, 5. The ends 2, 2, 3, 5 are all prime, so nothing can be split further: the tree is finished.
2. **C.** 9. 9 = 3 × 3, so it has more than two factors and is composite.
3. **A.** 5. 2 × 5 = 10 and 50 ÷ 10 = 5, so 50 = 2 × 5 × 5.
4. **D.** Both 6 and 8. Both 6 and 8 are composite: 6 = 2 × 3 and 8 = 2 × 4, and 4 still needs splitting.
5. **C.** 3 × 3 × 7. 63 = 7 × 9 and 9 = 3 × 3, so 63 = 3 × 3 × 7.

### Talk about it

Is every number at the bottom of the tree prime? How can you tell your tree is finished?

### Home mission

Write 84 on paper. Ask your student to start with 2 × 42 instead of 4 × 21. Do both trees end in the same primes?

### When they are stuck, ask

- Is every number at the bottom of the tree prime?
- Can this number be split into smaller factors?
- Do your primes multiply back to the number you started with?

### Try not to say

- Stop as soon as you have two factors.
- 1 counts as a prime factor.
- You have to start with the smallest factor or it will not work.